Fewer CX gates for exact unitary synthesis via single-qubit pivots

Matt Bowring
Purdue University
mbowring@purdue.edu

October 14, 2026

Abstract

This paper presents an improved quantum gate decomposition that results in smaller quantum circuits. The current best-known exact decomposition requires $O(\frac{22}{48}4^n)$ two-qubit gates for an arbitrary n-qubit unitary. I reduce this count to $O(\frac{21}{48}4^n)$ by introducing a parameterized single-qubit *pivot* gate that removes a two-qubit gate at each recursive step. The gate parameters are found by a one-dimensional root search rather than conventional gradient-based optimization. Their existence for generic unitaries is formally verified in `Lean` and demonstrated numerically on random instances. The practicality and potential improvements of the proposed decomposition are discussed.

I Introduction

I.A The synthesis problem

To run a quantum program on hardware, its unitary operations must be translated into the native gate set of the device, a task known as gate decomposition or synthesis. Given a target unitary matrix, the task is therefore to produce a quantum circuit using as few gates as possible. The common native gate set is $\{\mathrm{CX}\} \cup U(2)$, where CX is the two-qubit controlled-X (CNOT) gate and $U(2)$ is the group of single-qubit unitaries. A parameter count shows that an exact decomposition of a generic n-qubit unitary needs at least [1]

$$c_{\min}(n) \;\geq\; \left\lceil \tfrac{1}{4}(4^n - 3n - 1) \right\rceil$$

CX gates. These two-qubit gates are typically the bottleneck for quantum processors due to their connectivity constraints and dominant error rates.

I.B Existing work

The current approaches either provide an analytical decomposition or rely on numerical optimization of gate parameters. The best-known analytical methods [2, 3] follow the recursive pattern in Fig. 1. These have a direct algebraic formula but produce suboptimal circuits, whereas numerical approaches can reach near-optimal circuits at the cost of intensive gradient-based search (see, e.g., [4, 5]). The quantum Shannon decomposition (QSD) [2] introduced the recursive pattern using three multiplexed rotations (multiplexors) between four sub-unitaries, requiring $O(\frac{23}{48}4^n)$ CX gates. At each step, the multiplexors [6] apply a different rotation to the top qubit for each basis state of the control qubits, which effectively factor out the top qubit. Each multiplexor uses $d = 2^{n-1}$ CX gates for the bottom $n-1$ control qubits and dominates the overall CX count. The QSD was

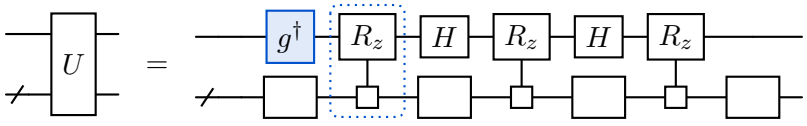


Figure 1: The template for decomposing a target n-qubit unitary U. A single-qubit pivot gate $g^\dagger$ (blue fill) allows the left-side multiplexor (blue dots) to use one fewer CX gate by merging it into the central multiplexor. This template is applied recursively to the four generic $(n-1)$-qubit unitaries (blank boxes), leading to a reduction of $(4^{n-2}-1)/3$ CX gates relative to Block-ZXZ.

further improved using the Block-ZXZ decomposition [3], which modified the central multiplexor to save one CX gate at each step, resulting in the current best-known count $O(\frac{22}{48}4^n)$ for exact decomposition.

I.C Contribution

I adapt the current best-known exact Block-ZXZ decomposition with a parameterized single-qubit gate at each recursive step, as depicted in Fig. 1 with blue highlights. This gate is chosen such that an additional CX gate can be merged into the central multiplexor, similar to the original technique used by Block-ZXZ to improve QSD. However, as outlined in Sec. II, the gate parameters do require optimization instead of the direct commutation rules used by Block-ZXZ. I obtain these parameters with a one-dimensional root search rather than gradient-based optimization. Section III presents numerical experiments and discusses the practicality and potential further improvements. Table 1 summarizes the main gate count result.

Number of qubits	1	2	3	4	5	6	n
QSD [2]	0	3	20	100	444	1868	$(23/48)\cdot 4^n - (3/2)\cdot 2^n + (4/3)$
Block-ZXZ [3]	0	3	19	95	423	1783	$(22/48)\cdot 4^n - (3/2)\cdot 2^n + (5/3)$
This work	**0**	**3**	**18**	**90**	**402**	**1698**	$\mathbf{(21/48)\cdot 4^n - (3/2)\cdot 2^n + 2}$
Theoretical lower bound	0	3	14	61	252	1020	$\lceil (4^n - 3n - 1)/4 \rceil$

Table 1: CX counts for exact synthesis of a generic n-qubit unitary.

II An improved decomposition

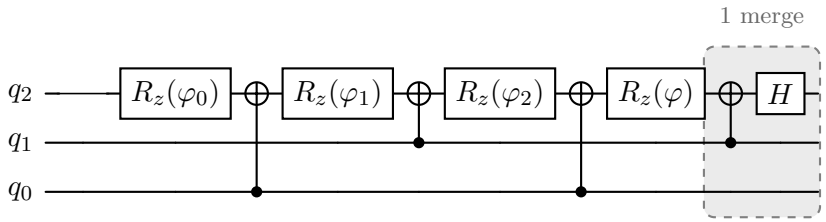
The results of this section are verified in `Lean` and assume the standard gate decompositions. While this paper omits the formal theorems and proofs, the `Lean` constructions are available [7]. Section II.A motivates the pivot gate, Sec. II.B states when the merge is possible, Sec. II.C shows how to compute the parameters, and finally Sec. II.D derives the main CX count result.

II.A Merging an additional CX gate

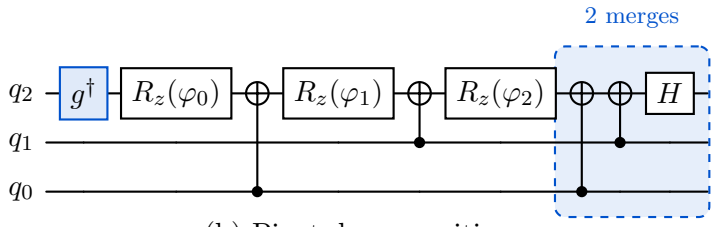
A generic multiplexed gate has a block-diagonal form $U_1 \oplus U_2$ with the well-known decomposition [2]:

$$U_1 \oplus U_2 = (I \otimes V)\, D(\boldsymbol{\theta})\, (I \otimes W), \qquad U_2 U_1^\dagger = V \,\mathrm{diag}(e^{i\theta_j}) V^\dagger, \tag{1}$$

where the diagonal matrix $D(\boldsymbol{\theta})$ is a multiplexed rotation, $\boldsymbol{\theta}$ are the eigenphases of $U_2U_1^\dagger$, and V, W are unitary matrices. Here $j = (j_{n-2}\cdots j_0)_2 \in \{0,\ldots,d-1\}$ labels the control basis states



(a) Standard decomposition



(b) Pivot decomposition

Figure 2: Decomposition of the left-side multiplexor with the pivot gate $g^\dagger$ (blue highlight of Fig. 1 with $n = 3$). (a) The final rotation $R_z(\varphi)$ allows only one CX gate (gray) to merge into the central multiplexor. (b) The pivot gate $g^\dagger$ makes $\varphi = 0$, so two CX gates (blue) can merge. The central multiplexor (not shown) uses the same number of CX gates after merging, so the left-side multiplexor costs 2 CX gates instead of 3.

in binary, with bit j_i the state of q_i. This multiplexed rotation is depicted in Fig. 2a, where the single-qubit rotation angles are computed by [6]

$$\boldsymbol{\varphi} = \frac{1}{d} H \, \boldsymbol{\theta}, \tag{2}$$

where H is the $d \times d$ Walsh–Hadamard matrix with rows in Gray-code order. As shown in Fig. 2b, the improvement comes from removing the final angle,

$$\varphi = \frac{1}{d} \mathbf{h}^\mathsf{T} \boldsymbol{\theta}, \tag{3}$$

where $\mathbf{h}$ is the row of H corresponding to the top control qubit q_{n-2}, i.e., $h_j = (-1)^{j_{n-2}}$. For example, Fig. 2 has $n = 3$, so $j = (j_1 j_0)_2$ runs over $|q_1 q_0\rangle = |00\rangle, |01\rangle, |10\rangle, |11\rangle$, and $\mathbf{h} = (1, 1, -1, -1)$. Removing the final rotation with $\varphi = 0$ requires that the eigenphases $\boldsymbol{\theta}$ be balanced with $\mathbf{h}$. To achieve this, the gate $g = R_z(\alpha) R_y(\beta)$ is inserted, giving the full decomposition

$$U = (*) \left(I \oplus C(g)\right) (g^\dagger \otimes I), \tag{4}$$

where $I \oplus C(g)$ is the left-side multiplexor and $(*)$ collects the remaining right-side gates. The gate g acts as a *pivot* by rotating the eigenspectrum of C to satisfy $\varphi = 0$. By (1), this modified left-side multiplexor can be decomposed as

$$C(g) = V \operatorname{diag}\left(e^{i\theta_0}, \ldots, e^{i\theta_{d-1}}\right) V^\dagger. \tag{5}$$

This decomposition is not unique. A common permutation of the eigenvalues and the columns of V leaves (5) unchanged, and each angle is fixed only modulo 2π, since a 2π shift flips the sign of one eigenvector, which V and W absorb. With $\mu_0, \ldots, \mu_{d-1}$ the eigenphases of $C(g)$,

$$\theta_j = \mu_{k(j)} + 2\pi m_j, \qquad m_j \in \mathbb{Z}, \tag{6}$$

for any assignment $j \mapsto k(j)$ of eigenphases to basis states.

II.B The balanced-split condition

By (3), φ depends only on how the eigenphases are assigned to the $+1$ and -1 entries of $\mathbf{h}$, and by (6) this assignment can be chosen. Let S, with $|S| = d/2$, be the set of eigenphases placed on the $+1$ entries. Then

$$\varphi = \frac{1}{d}\Big(\sum_{k\in S}\mu_k - \sum_{k\notin S}\mu_k\Big) + \frac{2\pi\ell}{d}, \qquad \ell \in \mathbb{Z}, \tag{7}$$

where $\ell = \sum_{h_j=+1} m_j - \sum_{h_j=-1} m_j$ is set by the branches and can take any integer value. Hence φ can be set to zero iff

$$\sum_{k\in S}\mu_k - \sum_{k\notin S}\mu_k = -2\pi\ell \qquad \text{for some } \ell \in \mathbb{Z}. \tag{8}$$

Such an S is a *balanced split*. For a fixed C this is nongeneric: the left side of (8) varies continuously with C, while the right side is discrete. The pivot, however, gives a continuous one-parameter family $C(\beta)$, so (8) becomes a scalar root condition in β.

II.C Finding the pivot angles

Let α^* and β^* denote the pivot angles that produce a balanced split. The angle β moves C from $C(0)$ to $C(0)^{-1}$, while α is chosen so that an ordered eigenphase list reverses and changes sign between these endpoints.

The path $C(\beta)$. Writing the target unitary as $U = \left(\begin{smallmatrix} X & Y \\ * & * \end{smallmatrix}\right)$, the pivoted top-row blocks become

$$X_g = e^{-i\alpha/2}\cos\tfrac{\beta}{2}\,X + e^{i\alpha/2}\sin\tfrac{\beta}{2}\,Y, \qquad Y_g = -e^{-i\alpha/2}\sin\tfrac{\beta}{2}\,X + e^{i\alpha/2}\cos\tfrac{\beta}{2}\,Y. \tag{9}$$

For fixed α, the Block-ZXZ factorization [3, Sec. 4.1] gives the multiplexor matrix of (4) in terms of these pivoted blocks:

$$C(\beta) = -i\,X_g^\dagger\,(X_gX_g^\dagger)^{-1/2}(Y_gY_g^\dagger)^{-1/2}\,Y_g, \tag{10}$$

where the inverse square roots are positive definite. The endpoint relation is

$$C(\pi) = C(0)^\dagger = C(0)^{-1}, \tag{11}$$

since at $\beta = \pi$ the blocks swap with a sign change, so the ending eigenphases are the negatives of the starting ones, modulo 2π and up to permutation. This symmetry yields a continuous real phase imbalance with opposite endpoint values, which therefore has a zero at some $\beta^* \in [0, \pi]$.

Choosing α. In (10) the inverse square roots have positive determinants and $(-i)^d = 1$, so $\det C$ has the phase of $\det Y_g / \det X_g$. With τ_k the eigenvalues of $e^{i\alpha}X^{-1}Y$, (9) gives

$$\frac{\det Y_g}{\det X_g} = \prod_k \left(\tau_k\,\frac{\cos\frac{\beta}{2} - \sin\frac{\beta}{2}/\tau_k}{\cos\frac{\beta}{2} + \tau_k\sin\frac{\beta}{2}}\right). \tag{12}$$

As β runs from 0 to π, the factor $\cos\frac{\beta}{2} + \tau_k\sin\frac{\beta}{2}$ moves from 1 to τ_k and $\cos\frac{\beta}{2} - \sin\frac{\beta}{2}/\tau_k$ from 1 to $-1/\tau_k$. For nonreal τ_k and $\beta > 0$, both lie in the open half-plane containing τ_k and cannot vanish. Thus choosing α with no real τ_k keeps X_g, Y_g invertible and $C(\beta)$ continuous. The two factors

accumulate phases $\arg\tau_k$ and $\operatorname{sgn}(\arg\tau_k)\,\pi - \arg\tau_k$, respectively. Subtracting the denominator phase from the numerator phase gives the total change in the continuous phase of $\det C$:

$$\pi\,(N_+ - N_-) - 2\sum_k \arg\tau_k, \qquad \arg\tau_k \in (-\pi, \pi), \tag{13}$$

where $N_\pm$ count the τ_k in the upper and lower half-planes. Choosing α^* with $N_+ = N_- = d/2$ removes the half-turn term. Varying α rotates all τ_k together. With distinct crossing angles, N_+ changes by one at each real-axis crossing; a rotation by π replaces N_+ by $d - N_+$. The count therefore reaches $d/2$ on an interval between crossings, and α^* is chosen in its interior.

Choosing β. With $\alpha = \alpha^*$, let $\sigma(\beta)$ be the continuous phase of $\det C(\beta)$ with $\sigma(0) = \sum_k \arg\tau_k$. By (13),

$$\sigma(\pi) = -\sigma(0). \tag{14}$$

Choose real representatives of the eigenphases of $C(\beta)$, ordered by rank k:

$$\nu_0(\beta) \le \cdots \le \nu_{d-1}(\beta) \le \nu_0(\beta) + 2\pi, \qquad \sum_k \nu_k(\beta) = \sigma(\beta). \tag{15}$$

The interval condition fixes the ordered list up to cyclic reordering with 2π branch shifts. Moving the smallest phase to the end after adding 2π increases the sum by 2π, so the sum condition selects a unique list. This choice varies continuously with β, including when a phase crosses the argument cut. Negating an increasing list reverses its order, so (11) and (14) give $\nu_k(\pi) = -\nu_{d-1-k}(0)$. The middle half $S = \{d/4, \ldots, 3d/4 - 1\}$ of the ranks is invariant under $k \mapsto d - 1 - k$, so the imbalance

$$P(\beta) = \sum_{k\in S} \nu_k(\beta) - \sum_{k\notin S} \nu_k(\beta) \tag{16}$$

satisfies

$$P(\pi) = -P(0), \tag{17}$$

and by continuity $P(\beta^*) = 0$ for some $\beta^* \in [0, \pi]$. Thus, taking $\mu_k = \nu_k(\beta^*)$, the middle half S satisfies (8) with $\ell = 0$ and placing these lifted eigenphases on the basis states with $h_j = +1$ and the rest on $h_j = -1$ gives $\varphi = 0$.

II.D CX gate count

Block-ZXZ needs at most $c_n \le 4c_{n-1} + 3\cdot 2^{n-1} - 5$ CX gates [3, Sec. 5.3]: four $(n-1)$-qubit children and three multiplexors of 2^{n-1} CX gates each, less one CX for each of the three children decomposed up to a diagonal [8] and one for each of the two CX gates merged into the central multiplexor. The pivot merges a third, so

$$c_n = 4c_{n-1} + 3\cdot 2^{n-1} - 6, \qquad c_2 = 3, \tag{18}$$

where $c_2 = 3$ is the optimal CX count for a generic two-qubit unitary [1]. Unrolling over the $(4^{n-2} - 1)/3$ recursion nodes as in [3, Sec. 5.3],

$$c_n = 4^{n-2}c_2 + 3\cdot 2^{n-1}\big(2^{n-2} - 1\big) - 2\big(4^{n-2} - 1\big) = \tfrac{21}{48}\,4^n - \tfrac{3}{2}\,2^n + 2.$$

Subtracting (18) from the Block-ZXZ recurrence, the saving $\Delta_n = c_n^{\mathrm{ZXZ}} - c_n$ satisfies $\Delta_n = 4\Delta_{n-1} + 1$ with $\Delta_2 = 0$, so

$$\Delta_n = \sum_{k=0}^{n-3} 4^k = \frac{4^{n-2} - 1}{3}, \tag{19}$$

one CX for each recursion node. This agrees with the difference of the closed forms, $\left(\frac{22}{48} - \frac{21}{48}\right)4^n + \frac{5}{3} - 2 = \frac{4^{n-2}-1}{3}$.

III Results and Discussion

Figure 3 compares the proposed pivot decomposition against the current best-known Block-ZXZ decomposition. Both methods are run on the same Haar-random unitaries. The target unitary matrix and reconstructed circuit matrix are equal to machine precision (up to global phase). All experiments were run on a MacBook Pro (M1 series, 2021) using standard linear algebra primitives from `numpy` and `scipy`.

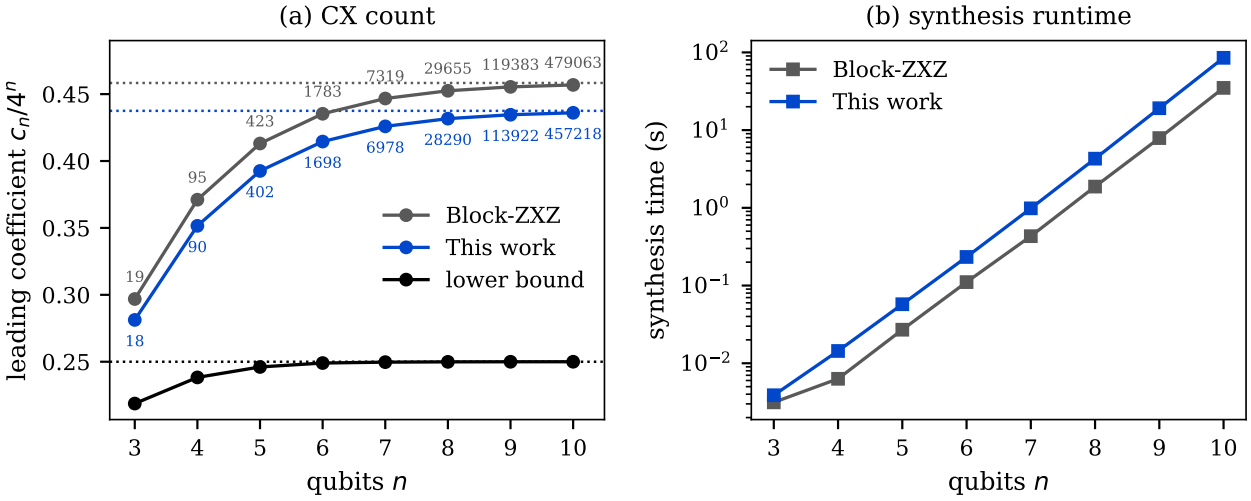


Figure 3: Comparison of the pivot decomposition (blue) with Block-ZXZ (gray). (a) The CX count divided by 4^n (solid dots) with the asymptotic coefficients (dotted lines). Each point is labeled with its CX count. (b) Total runtime of the decomposition.

The pivot decomposition produces quantum circuits using $(4^{n-2} - 1)/3$ fewer CX gates at the cost of added runtime. The counts reported in Fig. 3a match the expected recurrence formula.

Each recursive step already costs $O(d^3)$, and each root-search iteration also costs $O(d^3)$, so the overhead should be constant as long as the number of iterations does not grow with n. This depends on convergence tolerance and warrants further study: the empirical overhead in Fig. 3b is about 2–2.5×. The CX count is independent of the root-search method, so further runtime improvements may be possible.

A second pivot applied in reverse to the right-side multiplexor could perhaps save another $(4^{n-2} - 1)/3$ CX gates, although the two pivots are coupled and would require a joint search.

Acknowledgements

Under my supervision, Claude Opus 5.5 (Anthropic) was used to construct all `Lean` formalisms, generate the numerical experiments, and aid in writing this paper.

Code availability

The formal theorems and proofs of the results outlined in Sec. II and numerical experiments presented in Sec. III are available [7].

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